Supplementary Appendix to "Sticky Thresholds Under Vigilant Learning"

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This Supplementary Appendix presents the technical proofs, constants, and linearization details behind the paper's appendix, the full proof of the paper's finite-cost complete-episode proposition, a brief remark on possible vigilance-assisted exits, and computational material for the applications.

1 Quiescence near an E-stable RPE

We prove the three parts of Lemma 1. Together with the short argument in the paper's appendix they establish that, near an E-stable RPE, misspecification concern is small and V vanishes for small constant gains.

The regularity conditions as lemmas. The paper's Lipschitz condition could be derived rather than assumed. For the bound (16), the LLR's derivatives in its estimated arguments are polynomials in (x_t, x_{t-1}) scaled by the projected variance floors. Conditional on observed information the data are Gaussian with moments bounded on $\mathcal{Z}$, so the mean-value theorem delivers the bound with $\mathbb{E}L_t^2 \leq C$. The fluctuation order $\mathbb{E}\|\widehat{\vartheta}_t - \vartheta^*\|^2 = O(\gamma)$ of the benchmark estimates near an E-stable fixed point is the standard stochastic-approximation result (Evans and Honkapohja, 2001). The paper does not assume it. The proof of Lemma 1(iii) below derives it for the localized recursion. Phase 3 applies the Lipschitz bound at the pinned middle fixed point, intercept α_M, benchmark coefficients $(\alpha_M, 0)'$, and equal variances. In that regime, the PLM and benchmark predictive densities coincide pointwise, the LLR vanishes, and (16) simplifies to

$$\ell_t^+ \leq L_t\big(|c_t - \alpha_M| + \|\widehat{\vartheta}_t - \vartheta^M\|\big), \tag{1}$$

with $\widehat{\vartheta}_t$ the estimated benchmark coefficients entering the LLR and ϑ^M the associated values at the middle fixed point.

Proof of Lemma 1. Linearizing the constant-gain recursion for c_t and the actual law of motion produces

$$c_{t+1} - c^* = (1 + \gamma\lambda)(c_t - c^*) + \gamma\sigma\eta_t, \qquad x_t - c^* = \beta(c_t - c^*) + \sigma\eta_t + O_p(\gamma).$$

The stationary AR(1) variance is part (i). Since $\mathrm{Cov}(c_t, \eta_{t-1}) = \gamma\sigma$,

$$\mathrm{Cov}(x_t, x_{t-1}) = \gamma\sigma^2\left(\beta + \frac{\beta^2}{2|\lambda|}\right) + O(\gamma^2), \qquad \mathrm{Var}(x_t) = \sigma^2 + O(\gamma).$$

Dividing gives the lemma's stated correlation. Assumption 1 provides the sense of localization needed by the proposition.

For part (ii), let m and σ_y^2 denote the mean and variance of x_t, $y_t = x_t - m$, and define $\tilde{y}_t = {}^{y_t}/_{\sigma_y}$, $w_t = {}^{y_{t-1}}/_{\sigma_y}$, which are jointly standard normal with correlation ρ. Define $v_t = (\tilde{y}_t - \rho w_t)/\sqrt{1-\rho^2}$ as the (normalized) residual from the linear projection of $\tilde{y}$ on w, so that w_t, v_t are independent standard normals for each t (time subscripts suppressed below), and rearrange as $\tilde{y} = \rho w + \sqrt{1-\rho^2}\, v$. Substituting into the Gaussian likelihood ratio leads to

$$\ell^{\mathrm{pop}} = -\tfrac{1}{2}\log(1-\rho^2) + \frac{\rho^2}{2}(w^2 - v^2) + \rho\sqrt{1-\rho^2}\, wv.$$

Taking expectations leads to the expression for the mean. Since the $\max(\cdot, 0)$ is 1-Lipschitz, the last term determines the truncated mean to first order. Then we can use symmetry and $\mathbb{E}|w| = \sqrt{2/\pi}$ to show that $\mathbb{E}[(\rho w v)^+] = |\rho|/\pi$. The remaining terms have expected absolute value $O(\rho^2)$, proving part (ii).

Part (iii) is the constant-gain local-stability argument of Evans and Honkapohja (2001). The standard argument needs care for two reasons. First, the concern equation averages over the positive-part of the LLR, so its mean dynamics are Lipschitz but non-differentiable at the resting point. Second, the benchmark regresses on the lagged observation making the noise one-dependent. We verify stability with a Lyapunov function rather than a Jacobian.

Mean dynamics. Fix $\kappa < \infty$. Let p and q be the PLM and benchmark forecast variances, $\theta = (a, b)'$ the benchmark coefficients, and R the benchmark's moment matrix, each timed so that it is predetermined when x_t is observed. The slow state is $z = (c, \theta, R, p, q, \Lambda)$, and updating takes the form of a stochastic recursive algorithm $z_{t+1} = z_t + \gamma H(z_t, x_t, x_{t-1})$. With z held fixed, the actual law of motion makes the observations i.i.d. Gaussian, $x_t \sim N(m(z), \sigma^2)$, where $V(z) = \kappa\Lambda p$, and $m(z) = G(c) + \frac{1}{2}G''(c)V(z)$. Let $M(m)$ be the second-moment matrix of $(1, x_{t-1})'$ and $U(z) = \mathbb{E}_z[(\ell(z; x_t, x_{t-1}))^+]$ the mean of the truncated LLR conditional on holding z fixed. Averaging over (x_t, x_{t-1}) gives the mean-dynamics ODE

$$\begin{aligned} &\dot{c} = m - c, && \dot{\theta} = R^{-1}M(m)\{(m, 0)' - \theta\}, && \dot{R} = M(m) - R, \\ &\dot{p} = \sigma^2 + (m-c)^2 - p, && \dot{q} = \sigma^2(1+b^2) + (m - a - bm)^2 - q, && \dot{\Lambda} = U(z) - \Lambda. \end{aligned} \tag{2}$$

The θ equation is the recursive least squares ODE for the benchmark under the T-map $\theta \to (m(z), 0)'$. Holding z fixed there is no comovement so the benchmark slope coefficient is zero.

Rest point and linearization. Fix an outer-well $V = 0$ RPE $c_o \in \{\alpha_L, \alpha_H\}$. Then the mean-dynamics ODE (2) has resting point $z_o = (c_o, (c_o, 0)', M(c_o), \sigma^2, \sigma^2, 0)$. Define the differenced terms,

$$u = c - c_o, \qquad D = (a - c,\ b,\ q - p)', \qquad v = p - \sigma^2, \qquad H = R - M(c_o),$$

$\lambda_o = G'(c_o) - 1 = -6V^*/k < 0$, and $h_o = \frac{1}{2}\kappa\sigma^2 G''(c_o)$. A first-order approximation around z_o to $\dot{c}$ is $m - c = \lambda_o u + h_o\Lambda$. Since $R^{-1}M(m) = I + O(\|z - z_o\|)$, the coefficient equation

linearizes to $\dot{a} = (m - c_o) - (a - c_o)$ and $\dot{b} = -b$. The mean m enters $\dot{a}$ and $\dot{c}$ and cancels from $a - c$, while it enters as second-order (along with b^2) in the p, q equations. Hence

$$\begin{aligned} \dot{D} &= -D + O(\|z - z_o\|^2), \quad \dot{u} = \lambda_o u + h_o \Lambda + O(\|z - z_o\|^2), \\ \dot{v} &= -v + O(\|z - z_o\|^2), \quad \dot{H} = -H + M'(c_o)\{G'(c_o)u + h_o\Lambda\} + O(\|z - z_o\|^2). \end{aligned} \tag{3}$$

For the concern equation, the average is again over (x_t, x_{t-1}) at fixed z, but the positive part in U requires a pointwise bound on ℓ before averaging. Let $\tilde{z}$ be z replacing $(a, b, q) = (c, 0, p)$, so that z and $\tilde{z}$ differ by D. At $\tilde{z}$ the PLM and benchmark predictive densities are the same normal density, so $\ell(\tilde{z}; x_t, x_{t-1}) = 0$ for every realization. Assumption 2 is a Lipschitz bound in the estimated arguments with a data-dependent constant, so $|\ell(z; x_t, x_{t-1})| \le L_t \|D\|$ for every realization, and then averaging at fixed z leads to

$$0 \le U(z) \le C_U \|D\|, \qquad C_U = \sup_{\mathcal{Z}} \mathbb{E} L_t < \infty. \tag{4}$$

Local exponential stability. Equations (3)–(4) are block recursive. The model differences evolve independently and D drives concern Λ which drives intercepts u, then the moments H by u and Λ. Each block has a negative diagonal, and v is uncoupled. Since U is not differentiable at $D = 0$, we prove stability with the weighted quadratic

$$Q(z - z_o) = w_D \|D\|^2 + w_\Lambda \Lambda^2 + w_u u^2 + w_v v^2 + w_H \|H\|_F^2.$$

With positive weights declining enough down the recursive ordering, there is an $\omega > 0$ and a neighborhood of z_o, both depending on κ but not on γ, on which

$$\nabla Q \cdot h(z) \le -\omega\, Q(z - z_o), \tag{5}$$

where h is the right-hand side of (2). The reason is that each block of (3) depends only on blocks prior, so once the previous weight is big enough Young's inequality dominates every cross term by the negative diagonal terms of the two blocks connected by it, and the quadratic remainders are $O(Q^{3/2})$.

Fluctuations. Following the projection facility of Evans and Honkapohja (2001), fix $0 < c_1 < c_2$ with $\{Q \le c_2\}$ inside the neighborhood of (5). The learning rule is followed while z_t stays in $\{Q \le c_2\}$ and is otherwise projected to a point of $\{Q \le c_1\}$. The projection does not increase Q, and the projected recursion coincides with the actual until the estimates first leave $\{Q \le c_2\}$. The local stationary regime of the lemma is the stationary distribution of this projected recursion. Its step-increments satisfy $\mathbb{E}\|z_{t+1} - z_t\|^2 \le C\gamma^2$, because Assumptions 1 and 2 bound the second moments of x_t^2, ℓ_t^+, and R_{t+1}^{-1} on the set. Given the information preceding x_{t-1}, that observation and x_t are fresh Gaussian draws with means set by z_{t-1} up to $O(\gamma)$, so the conditional mean of the increment is $\gamma h(z_{t-1}) + O(\gamma^2)$. Evaluating ∇Q one step back as well costs a further $O(\gamma^2)$, and (5) then gives, after decreasing ω,

$$\mathbb{E} Q(z_{t+1} - z_o) - \mathbb{E} Q(z_t - z_o) \le -\omega\gamma\, \mathbb{E} Q(z_t - z_o) + C\gamma^2. \tag{6}$$

In the local stationary regime the left side vanishes, so $\mathbb{E} Q(z_t - z_o) \le C\gamma/\omega$ and we arrive at $\mathbb{E}\|z_t - z_o\|^2 \le C\gamma$.

The estimated LLR. Finally, $|\ell_t^{\text{est}}| \leq L_t\|D_t\|$ by the same comparison behind (4), but now with the real-time estimates. Cauchy–Schwarz and Assumption 2 give

$$\mathbb{E}|\ell_t^{\text{est}}| + \mathbb{E}(\ell_t^{\text{est}})^+ \leq 2\,(\mathbb{E}L_t^2)^{1/2}(\mathbb{E}\|D_t\|^2)^{1/2} \leq C\sqrt{\gamma},$$

which is part (iii). □

2 Transition detectability

This subsection contains the supporting results and proofs for Section 4.4.

Proof of Lemma 2. For part (i), admissibility implies that $\dot{\psi} = \psi - G(\psi)$ and so $\xi_\psi = \psi + \dot{\psi} = H(\psi)$. The chain rule leads to the $L_\xi = L_H L_G$ Lipschitz bound. Since the cubic fixed points are also fixed points of H, the truncated transitional endpoints $\psi(0)$ and $\psi(\Delta\tau)$ are within $L_H\varepsilon$ of α_H (or α_L) and α_M. It follows then that the endpoint displacement must be at least $D_\xi^{\min} \equiv \Delta/2 - 2L_H\varepsilon > 0$, where the inequality comes from $\varepsilon < \Delta/(4L_H)$. Finally, $H' = 2 - G'$ is quadratic, so it has at most two zeros. Then since ψ is monotonic there are no more than three monotone segments.

We prove part (ii), which is non-standard and specific to the model's cubic map and its intercept-only-PLM-versus-AR(1)-benchmark specification, in two steps. The first step uses a pigeonhole argument that exploits the cubic map. The second step is specific to the forecasting-model pair. Splitting the transition into its burn-in and post-burn-in portions, the burn-in can move the signal by no more than $L_\xi\tau_a$. So after burn-in, the displacement is at least $R_\xi = D_\xi^{\min} - L_\xi\tau_a > 0$. Then, by pigeonholing over the three monotone stretches, there is a subwindow $J_\psi = [a, a+h]$, of fixed length h, on which ξ_ψ moves monotonically by at least some $d_0 > 0$, uniformly in ψ. Evidently, one monotone stretch must account for at least $R_\xi/3$ of the post-burn-in displacement, so the length of that monotone interval is bounded below by $R_\xi/3L_\xi$. Since R_ξ, L_ξ, and the number of monotone intervals emerge as constants of the admissible class (Definition 3), h and d_0 depend only on $\mathcal{A}$, not on the exact transition path ψ.

Now focus on the window J_ψ. For $\lambda \in [0,1]$, monotonicity delivers a first time τ_λ at which the signal has covered the fraction λ of its displacement d_0. Take the blocks $A \equiv \left[\tau_{1/4}, \tau_{3/8}\right]$ and $B \equiv \left[\tau_{5/8}, \tau_{3/4}\right]$, whose signal values are separated by at least $d_0/4$, and the later block $K \equiv \left[\tau_{7/8}, a+h\right]$ of length at least $\varphi \equiv d_0/(8L_\xi)$. Evaluating the exponential window weights (normalized to integrate to one) at every $s \in K$ gives accumulated weights on A and B of at least a constant $p_0 > 0$. This follows because Lipschitzness makes each block length at least $d_0/(8L_\xi)$, and the weights are bounded above and below on the fixed horizon. The two-point variance identity then provides a lower bound for the weighted signal variance

$$\text{Var}_s^w(\xi_\psi) \geq (d_0/4)^2 p_0^2 =: v_E > 0.$$

Lipschitzness puts a bound $L_\xi\gamma$ on how much the signal can move over a single lag. Then applying Cauchy–Schwarz, for small γ, implies $\text{Cov}_s^w(\xi_\psi(r), \xi_\psi(r-\gamma)) \geq v_E/2$. Adding independent Gaussian shocks does not change this covariance, while projection and Gaussian moments bound the variances. The weighted-population AR(1) slope, the ratio of the

weighted lag-one covariance to the weighted lag variance, is therefore bounded below on K,

$$b_E(s;\psi) \geq \underline{b}_E > 0.$$

A nonzero slope makes the two predictive densities distinct. The LLR at s is a function of $(X_s, X_{s-\gamma})$, the current observation and its lag, which equals $(\xi_\psi(s)+\sigma\eta_s,\ \xi_\psi(s-\gamma)+\sigma\eta_{s-\gamma})$ with independent standard normal shocks. The benchmark's conditional mean varies with the lagged observation while the intercept-only mean does not, so except for a single value of $X_{s-\gamma}$ the two Gaussian densities differ. Further, the set on which the benchmark density is strictly larger contains a nonempty open set. Since $(X_s, X_{s-\gamma})$ has a nondegenerate joint Gaussian distribution, we have the event $\{\ell_E > 0\}$ with positive probability and, therefore, $\mathbb{E}(\ell_E^+) > 0$, uniformly. Then the truncated $\mathbb{E}(\ell_E^+)$ is continuous in the benchmark coefficients, the two variance values, and the means of the pair, follows from the dominated convergence theorem with a Gaussian distribution function. These parameters range over fixed compact sets, projection for the estimated quantities and the truncated admissible intervals for X, and the slope constraint $b \geq \underline{b}_E$ is closed. A continuous positive function on a compact set has a positive minimum and so,

$$u^E(s;\psi) = \mathbb{E}(\ell_E^+) \geq u_E > 0 \qquad \text{on } K, \text{ uniformly over } \psi.$$

The bound depends on the truncation. The benchmark doesn't have to fit the current pair $(X_s, X_{s-\gamma})$ on average it just needs the non-zero slope coefficient to makes its predictions detectably different so that vigilance then accumulates that difference (when positive).

It remains to transfer this population bound to the mean dynamics. Every moment that is part of the benchmark model is an exponentially weighted average, so we can focus on a generic moment recursion,

$$\bar{M}_{n+1} = (1-\gamma)\bar{M}_n + \gamma m(n\gamma).$$

Iterating, at $s = n\gamma$,

$$\bar{M}_n = (1-\gamma)^n \bar{M}_0 + \gamma \sum_{j<n} (1-\gamma)^{n-1-j} m(j\gamma).$$

The first term differs from $e^{-s}\bar{M}_0$ by $O(\gamma)$ on a fixed horizon, and Lipschitzness of the signal ξ_ψ makes the sum just an $O(\gamma)$ Riemann error away from the exponentially weighted population integral. After the transient period, the distance from the population moment is therefore bounded by

$$C\{e^{-\tau_a} + \gamma\}.$$

The first term in the bound comes from the decaying weight on the quiescent initial condition. The second term comes from the Riemann approximation error. Since the projected moment matrix is uniformly nonsingular, the coefficients inherit the same bound. The remaining component is the recursive variance estimate, an exponentially weighted average of squared PLM-residuals, its bound carries the same two terms as above and then also a term coming from PLM-coefficients differing from their population values. Then the bound follows since the mean-dynamics and population residuals are derived from model coefficients that differ

by less than the derived bound. Assumption 2 and the 1-Lipschitz positive-part map then give

$$\sup_{s\in J_\psi} |\bar{u}(s;\psi) - u^E(s;\psi)| \le C\{e^{-\tau_a} + \gamma\}.$$

Choosing τ_a^* and then γ so that errors are smaller than $u_E/2$, and integrating over K, leads to

$$\int_{J_\psi} \bar{u}(s;\psi)\, ds \ge \frac{\varphi u_E}{2} = h u_0, \qquad u_0 \equiv \frac{\varphi u_E}{2h} > 0,$$

which proves part (ii) with a constant that is independent of the transition path.

For part (iii), the standard weak-convergence theorem for constant-gain recursions (Evans and Honkapohja, 2001, Proposition 7.8) does not directly apply. Along a transition the observed process has a time-varying mean, so the state dynamics are not time-homogeneous, and the benchmark's lagged regressor makes the coefficients' updating one-dependent. Instead, center the update at the deterministic mean-dynamics path and then a direct finite-horizon argument delivers the $O_p(\sqrt{\gamma})$ rate. Assumption 1 again gives compactness, and Assumption 2 supplies uniform local Lipschitz and second-moment bounds. With the mean-dynamics path remaining in the nonbinding interior of the projection region, the projection facility never binds on this window and its term can be dropped. Center the update at the deterministic state,

$$M_{j+1} \equiv F_Z(\bar{Z}_j, X_j^\psi, X_{j-1}^\psi) - \mathbb{E}F_Z(\bar{Z}_j, X_j^\psi, X_{j-1}^\psi).$$

This variable is centered, depends only on (η_j, η_{j-1}), and has uniformly bounded second moment. The delicate step is bounding the noise sum uniformly over the window, which is what the Grönwall argument below requires. Because consecutive terms share an innovation, no choice of filtration makes the full sum a martingale, and Doob's maximal inequality does not apply directly. Instead, split the sum into its even- and odd-indexed terms. The even-indexed terms are independent of each other, similarly for the odd-indexed terms, so the two sums is a martingale and Doob's inequality applies to each separately.[1] Hence, for every $K_0 > 0$,

$$\Pr\left(\sup_{m\le N}\left\|\gamma\sum_{j<m} M_{j+1}\right\| > K_0\sqrt{\gamma}\right) \le \frac{C}{K_0^2},$$

uniformly in ψ. This is the required $O_p(\sqrt{\gamma})$ noise bound. The remaining discrepancy comes from approximating the (continuous time) mean-dynamics with its one-step (Euler) discretization that, since the difference is deterministic, leaves a $O(\gamma^2)$ remainder per period. Then accumulates at $O(\gamma)$ over the fixed window. Assumption 2 also gives $\gamma\sum_{j<N} L_j = O_p(1)$. Subtracting the mean Euler recursion from the stochastic recursion and applying the discrete Grönwall inequality then leads to the stated uniform tracking bound, including the assumed $O_p(\sqrt{\gamma})$ initial gap.

For the LLR claim, center $u(\bar{Z}_j, X_j^\psi, X_{j-1}^\psi)$ at its unconditional mean $\bar{u}(j\gamma;\psi)$, apply the exponentially weighted average, and then use the same even-odd split as above. This leaves a weighted sum $O_p(\sqrt{\gamma})$. Then the difference from replacing $\bar{Z}_j$ with $Z_j^{\gamma,\psi}$ is bounded by the

[1] Note that the first increment includes the pre-transition lag and the initial random shock. These, though, are multiplied by γ and so only contribute $O_p(\gamma)$, so we drop those terms.

tracking error times $\gamma \sum_j (1-\gamma)^{N-1-j} L_j$. Using Cauchy–Schwarz and Assumption 2 make this $o_p(1)$. □

Proof of Proposition 5. Fix $\psi \in \mathcal{A}$. Lemma 2(ii) finds a sub-window, following a burn-in period, $J_\psi = [a, a+h] \subset [\tau_a, \Delta\tau]$ that features monotone ξ_ψ path that moves at least d_0. Lemma 2(ii) finds that for $\tau_a \geq \tau_a^*$, $\int_{J_\psi} \bar{u}(s;\psi)\, ds \geq h\, u_0$ for small γ. Since $e^{-(\Delta\tau - s)} \geq e^{-\Delta\tau}$ on $[0, \Delta\tau]$ and $\bar{u} \geq 0$,

$$\ell_*(\psi) = \int_0^{\Delta\tau} e^{-(\Delta\tau - s)} \bar{u}(s;\psi)\, ds \geq e^{-\Delta\tau} \int_0^{\Delta\tau} \bar{u}(s;\psi)\, ds \geq e^{-\Delta\tau} \int_{J_\psi} \bar{u}(s;\psi)\, ds \geq e^{-\Delta\tau}\, h\, u_0.$$

The bound applies uniformly in ψ. □

3 Collapse and lingering

This subsection proves the Phase 3 results of Section 4.5.

The economics is straightforward. Above the threshold, the flattened landscape makes α_M an attractor and beliefs are pinned there. But at α_M the AR(1) benchmark and the intercept-only PLM make identical population predictions (F3) so the accumulated evidence of misspecification that raised vigilance and created the episode decays at slow-time rate one. When V recrosses V^*, the landscape reforms, α_M is again E-unstable and turns into a repeller as the linearization rate $\lambda(V) = (3/k)(V^* - V)$ of (F2) passes smoothly through zero and shocks force an escape in slow-time proportional to $\log(1/\gamma)$. The phase 3 then consists of three subphases: pinning at α_M, declining vigilance, and then release from the lingering near α_M and re-entry into a stable outer well.

While Phase 3 is easy to describe, the proofs are involved because each substage involves an actual crossing time instead of a deterministic date. We define the threshold-crossing times τ_+, τ_- of (19) as the realized periods that mark entry into and exit from the pitchfork neighborhood $V \in [V^* - \nu, V^* + \nu]$, and the durations are endogenous. The analysis traces how the window's path contracts the deviation period by period for an actual realized vigilance path, however long the window turns out to be. The proof below treats one substage per part.

It is useful to define two objects that play recurring roles in the Phase 3 proofs. The first packages the cubic landscape from the proof of Proposition 1 into the learning recursion. By that computation the deviation $\delta_t \equiv c_t - \alpha_M$ has the exact drift $\mu(\delta_t; V_t) = \lambda(V_t)\, \delta_t - \delta_t^3/k$ with $\lambda(V) = \frac{3}{k}(V^* - V)$, so with $x_t = T_{V_t}(c_t) + \sigma\eta_t$, the constant-gain update $c_{t+1} = c_t + \gamma(x_t - c_t)$ factors as

$$\delta_{t+1} \;=\; g_t\, \delta_t \;+\; \gamma\sigma\, \eta_t, \qquad g_t \equiv 1 + \gamma\lambda(V_t) - \gamma\, \delta_t^2/k. \tag{7}$$

On the projected state space (Assumption 1), $|\delta_t|$ and V_t are bounded, so there is $\gamma_0 > 0$ such that $g_t > 0$ for all $\gamma < \gamma_0$; and since the cubic term has the opposite sign of δ_t its impact is contracting and so we have the one-sided bound

$$g_t \;\leq\; 1 - \gamma\, a_t, \qquad a_t \equiv \frac{3}{k}\{V_t - V^*\}_+,$$

where a_t is the positive part of the contraction rate. On the strict high-vigilance window $[\tau_0, \tau_+)$ the rate satisfies $a_t > \lambda^* \equiv 3\nu/k$ by the definition of τ_+, and its slow-time integral over that window is the vigilance budget $A^-(\gamma)$ of (20). The second object is the release scale when the system exits the pitchfork neighborhood,

$$D_\gamma \equiv |c(\tau_-) - \alpha_M| \vee \sqrt{\gamma}.$$

Proof of Lemma 3. On $[\tau_0, \tau_+)$, $V_t > V^* + \nu$, so $0 < g_t \leq 1 - \gamma\lambda^*$. Iterating (7), the term capturing movement from the initial state is less than

$$\exp\left\{-\gamma \sum_{t_0 \leq j < t_+} \frac{3}{k}(V_j - V^*)\right\} = e^{-A^-(\gamma)+O(\gamma)}.$$

This expression captures that the amount of slack depends on the accumulation of excess vigilance, the budget $A^-(\gamma)$. The iteration of (7) yields a recursive expression for the moving-average terms whose accumulated variance is $O(\gamma)$, which provides an $O_p(\sqrt{\gamma})$ noise floor. So, iterating up to the random crossing time τ_+, taking the larger of the two bounds, proves part (i).

For part (ii), the claim is that beliefs enter a neighborhood around α_M quickly and, for small γ and ν, the path does not reach the boundary $C_{\text{tube}}\epsilon_0$ before τ_-, which marks entrance into a phase where the E-stable outer wells are restored. Entry into the tube follows from the uniform slack: the mean path reaches the $\epsilon_0/2$-neighborhood of α_M within the bounded time τ_{flow} and its Jensen intensity is still above $V^* + \nu$. Then repeating the same fixed-horizon tracking argument in Lemma 2(iii) at τ_0 puts the realized belief inside the ϵ_0-neighborhood at $\tau_0 + \tau_{\text{flow}}$, with V still above $V^* + \nu$, with probability tending to one. Lingering in this region comes from two facts proven below. First, so long as $V \geq V^* + \nu$ the factorization (7) contracts, implying that c's deviation from α_M cannot grow beyond the pinning at ϵ_0 plus shocks. Second, the time spent traversing the pitchfork neighborhood $[V^* - \nu, V^* + \nu]$ is too brief to measurably amplify the deviation. At $c = \alpha_M$, $X_t = \alpha_M + \sigma\eta_t$ is iid and the population AR(1) coefficients are $(\alpha_M, 0)$. The benchmark and PLM predictive densities are identical, so we have $\ell \equiv 0$ pointwise. The pointwise identity is stronger than $\mathbb{E}\ell = 0$, and is what drives $\mathbb{E}\ell^+ = 0$.

Now consider times after $\tau_0 + \tau_{\text{flow}}$, with the lingering process stopped the first time the system exits the $C_{\text{tube}}\epsilon_0$ neighborhood. On this stopped process, the benchmark coefficients arrive at $\tau_0 + \tau_{\text{flow}}$ still carrying their transition estimates, with $b_c \neq 0$ because of the detected Phase 2 transition trend. With c pinned at α_M, the observations are approximately i.i.d. and the benchmark's constant-gain estimates adjust toward the population coefficients $(\alpha_M, 0)$. This gap between the actual coefficients and their population values declines like e^{-s}. The tracking argument in Lemma 2(iii) plus Assumption 2 transfers the population identity $\ell \equiv 0$ into the bound

$$\bar{u}_M^\gamma(\tau) \leq C\{C_{\text{tube}}\epsilon_0 + e^{-(\tau - \tau_0 - \tau_{\text{flow}})}\} + O_p(\sqrt{\gamma}).$$

Variation of constants in (7) brings together two decaying exponentials, one from the initial concern and the other from the benchmark's coefficient gap at entry into the lingering process. These give the se^{-s} term in the bound on Λ. Thus, vigilance declines exponentially toward a floor of order $CC_{\text{tube}}\epsilon_0 + O_p(\sqrt{\gamma})$. Then multiplying the vigilance concern bound by $\kappa\bar{\sigma}^2$,

using the projection bound $\widehat{\sigma}^2 \le \overline{\sigma}^2$, the Jensen intensity V declines exponentially toward a floor of $\kappa\overline{\sigma}^2 C C_{\text{tube}}\epsilon_0 + O_p(\sqrt{\gamma})$. The lemma chooses ϵ_0 so that the Jensen intensity's floor is no greater than $\frac{1}{2}(V^* - \nu)$, a value that is strictly below the threshold for the release from the pitchfork neighborhood. The downward crossing, then, occurs within a fixed slow time, which gives the bound $\overline{T}$ on the time until release from the pitchfork neighborhood, $\tau_- - \tau_0$. The release-time bound depends only on the projection ceiling and the decay rate, not on ϵ_0. The lemma's constants are fixed in order: the release-time bound $\overline{T}$ comes first, determined by the projection ceiling and the decay rate alone; the lingering tube constant $C_{\text{tube}} > 2e^{\lambda^*\overline{T}}$ is chosen so that the tube survives even the worst-case amplification over a time until release of length $\overline{T}$; the ϵ_0 then satisfies the two conditions in the statement; and finally γ is taken small. Finally, the repelling force can amplify the entry displacement (smaller than ϵ_0) by no more than the factor $e^{\lambda^*\overline{T}} < C_{\text{tube}}/2$, which together with the $O_p(\sqrt{\gamma})$ shocks keeps the belief c strictly inside the tube. The path for c cannot reach the tube boundary before V exits the pitchfork neighborhood at τ_-. The continuation argument stops at that first-exit time.

For part (iii), iterate the factorization (7) across the pitchfork crossing $[\tau_+, \tau_-]$. During the crossing, $\lambda(V_t) \le \lambda^*$, so the product of the factors g_t multiplies the displacement at τ_+ by no more than $e^{\lambda^*(\tau_- - \tau_+)}$, a bounded factor because part (ii) makes the crossing time $O_p(1)$. The shocks accumulated over the crossing are $O_p(\sqrt{\gamma})$, by the same conditional-variance argument as in part (i). So then,

$$|c(\tau_-) - \alpha_M| \le e^{\lambda^*(\tau_- - \tau_+)}\,|c(\tau_+) - \alpha_M| + O_p(\sqrt{\gamma}),$$

and combining with part (i) and the definition of the release scale D_γ leads to $D_\gamma = O_p(e^{-A^-(\gamma)} \vee \sqrt{\gamma})$.

It remains to prove part (iv) and the two escape-from-lingering bounds. The logic is straightforward. Once the system leaves the pitchfork neighborhood, the highest (local) repulsion rate from α_M is $3V^*/k$. This escape force can generate a drift that renews vigilance in a way that temporarily flattens the T-map possibly making α_M attracting again. But, this creates a subtlety since every episode like this takes away from the cumulative repulsion from α_M. The argument shows that the cumulative stabilizing vigilance force is bounded and any renewed concern only delays the exit by a bounded amount that enters the relevant repulsion exponent.

First reset time indices relative to t_- and define

$$a \equiv \frac{3V^*}{k}, \qquad \delta_j \equiv c_{t_-+j} - \alpha_M, \qquad V_j \equiv V_{t_-+j}.$$

The law of motion for δ is

$$\delta_{j+1} = (1 + \gamma b_j)\delta_j + \gamma\sigma\eta_{t_-+j}, \qquad b_j \equiv a - \frac{3}{k}V_j - \frac{1}{k}\delta_j^2.$$

where from earlier

$$P_n \equiv \prod_{j<n}(1 + \gamma b_j), \qquad S_n \equiv P_n^{-1}\delta_n.$$

Now choose the fixed escape radius, ε_{esc} small enough that $a_\varepsilon \equiv a - \frac{\varepsilon_{\text{esc}}^2}{k} > 0$. Projection uniformly bounds b_j so $1 + \gamma b_j > 0$ for small γ. Iterating the recursion leads to

$$S_n = \delta_0 + \gamma\sigma \sum_{m<n} P_{m+1}^{-1} \eta_{t_-+m} \equiv \delta_0 + M_n. \tag{8}$$

Because P_{m+1}^{-1} is measurable before η_{t_-+m} is realized, M_n is a martingale. The first term in (8) is the initial distance $c - \alpha_M$ at the moment of release from the pitchfork ($V < V^* - \nu$). The second term is the new learning noise generated after release.

A stopping-time argument. The proof proceeds by a stopping-time argument and the key object here is a bounding condition on the cumulative repulsion that can be called Condition P. For $n \geq 0$, define the accumulated amount that any vigilance on the window drags against the repulsion, and call it the exit-budget

$$A_n^+ \equiv \frac{3\gamma}{k} \sum_{j<n} V_j,$$

This measures the cumulative Jensen adjustment that works to push against escape. For a constant $B > 0$ to be chosen below, let

$$\begin{aligned} N_{\text{esc}} &\equiv \inf\{n \geq 0 : |\delta_n| \geq \varepsilon_{\text{esc}}\}, \\ N_B &\equiv \inf\{n \geq 0 : A_n^+ \geq B\}, \\ N_T &\equiv \left\lfloor \frac{\frac{1}{2}\log(1/\gamma) + C_T}{a_\varepsilon \gamma} \right\rfloor, \qquad \bar{N} \equiv N_{\text{esc}} \wedge N_B \wedge N_T. \end{aligned}$$

N_{esc} is the escape time (object of interest), N_B is the first time the exit budget hits its bound, and N_T is a deterministic time that describes the mechanical expansion from an initial state of order $\sqrt{\gamma}$, going up to $\sqrt{\gamma}\, e^{a_\varepsilon \gamma N_T} \approx e^{C_T}$. The proof first operates on this stopped interval $[0, \bar{N}]$ and then shows that, with probability approaching one, the binding stop-time is N_{esc}. The order of constants is important. First is to set a preliminary radius $\bar{\varepsilon}$ with $a_{\bar{\varepsilon}} > 0$, then the constant C_{inh}, and the exit-budget bound $\widehat{B}$, all independent of the initial-block L. For a prescribed error probability choose, in order, K, r_0, L. Next pick $\varepsilon_{\text{esc}}^0 \leq \bar{\varepsilon}$ small enough to absorb the occupation and likelihood-ratio contributions, then choose C_T, and finally take γ small. This order makes all of the estimates uniform for $0 < \varepsilon_{\text{esc}} \leq \varepsilon_{\text{esc}}^0$ and allows $e^{2\widehat{B}} e^{-2a_{\bar{\varepsilon}} L}$ to be arbitrarily small by increasing L.

For every $0 \leq m < n \leq \bar{N}$, the expanding of $\log(1 + \gamma b_j)$ gives

$$\log \frac{P_n}{P_m} = a\gamma(n - m) - \frac{3\gamma}{k} \sum_{j=m}^{n-1} V_j - \frac{\gamma}{k} \sum_{j=m}^{n-1} \delta_j^2 + o(1). \tag{9}$$

The higher-order remainder term, $\sum_{m \leq j < n} |\log(1 + \gamma b_j) - b_j|$, is uniform because the projection facility implies that $|\log(1 + \gamma b_j) - \gamma b_j| \leq C\gamma^2$, with C depending on the $|b_j|$ uniform upper-bound. Before an escape, $|\delta_j| < \varepsilon_{\text{esc}}$. Before a budget-induced stop, N_B, every subinterval has no more than the budget B pushing against escape, up to a one-period overshoot

that is proportional to the small gain. Then (9) implies Condition P, the key bound in the proof:

$$e^{a_\varepsilon \gamma(n-m)-B-o(1)} \le \frac{P_n}{P_m} \le e^{a\gamma(n-m)+o(1)}. \tag{10}$$

This bound on cumulative repulsion permits V_j to recross V^*, so $\lambda(V) < 0$ is possible but the cumulative amount of contracting back towards α_M is sufficiently small to not interrupt escape.

Claim 1: learning noise displaces beliefs from the threshold, the repulsion amplifies that displacement faster than subsequent noise can reverse it, and the path stays far from the threshold only briefly. For every fixed $B < \infty$, on events that can be made arbitrarily close to probability one for sufficiently small γ, there are two cases. If the stopped interval ends during an initial fixed slow-time interval of length L, then the occupation bounds below are immediate. If the process survives that interval, there is a date $n_0 \le N_L = \lfloor L/\gamma \rfloor$ after which S_n keeps its sign, remains bounded away from zero at the $\sqrt{\gamma}$ scale, and stays within a fixed multiple of its magnitude. In particular,

$$S_j S_n > 0, \qquad C_S^{-1} \le \frac{|S_j|}{|S_n|} \le C_S, \qquad |S_n| \ge r_0\sqrt{\gamma}, \qquad n_0 \le j, n \le \bar{N}, \tag{11}$$

for fixed constants $C_S < \infty$ and $r_0 > 0$. In either of the cases,

$$\gamma \sum_{j<\bar{N}} |\delta_j| \le C_B \varepsilon_{\text{esc}} + o_p(1), \qquad \gamma \sum_{j<\bar{N}} \delta_j^2 \le C_B \varepsilon_{\text{esc}}^2 + o_p(1), \tag{12}$$

where $C_B < \infty$ depends on B but not γ.

To prove the claim, first notice from (10) that, on $[t_-, \bar{N}]$,

$$P_{m+1}^{-1} \le e^{B+o(1)} e^{-a_\varepsilon(m+1)\gamma}.$$

For the rest of this proof, M_n is interpreted to be the martingale on the stopped interval $n < \bar{N}$, where $M_n = \gamma\sigma \sum_{m<n} P_{m+1}^{-1} \eta_{t_-+m} \mathbf{1}_{m<\bar{N}}$, with the Condition P and Doob's inequality implying that

$$\mathbb{E} \sup_{n \le \bar{N}} |M_n|^2 \le C\gamma^2 e^{2B} \sum_{m \ge 0} e^{-2a_\varepsilon(m+1)\gamma} \le C_B \gamma. \tag{13}$$

For the claim on S_n there are 2 cases depending on whether the post t_- value δ_0 is above or below the $O(\sqrt{\gamma})$ learning noise floor. Let $|\delta_0| \ge K\sqrt{\gamma}$. From Doob's inequality, $\Pr\left(\sup_{n<\bar{N}} |M_n| \ge \frac{1}{2}|\delta_0|, |\delta_0| \ge K\sqrt{\gamma}\right) \le \Pr\left(\sup_{n<\bar{N}} |M_n| \ge {}^{K}/_{2}\sqrt{\gamma}\right) \le {}^{4C_B}/_{K^2}$. Choosing K large ensures $|M_n| < {}^{1}/_{2}|\delta_0|$ outside a small probability event. The reverse triangle inequality implies $|S_n| \ge |\delta_0| - |M_n| \ge {}^{1}/_{2}|\delta_0|$. The triangle inequality implies that $|S_n| \le |\delta_0| + |M_n| \le {}^{3}/_{2}|\delta_0|$. Hence ${}^{1}/_{2}|\delta_0| \le |S_n| \le {}^{3}/_{2}|\delta_0|$. S_n preserves δ_0's sign and $|S_n| \ge {}^{K}/_{2}\sqrt{\gamma}$.

Now let $|\delta_0| < K\sqrt{\gamma}$. Fix a slow time $L > 0$ and denote $N_L = \lfloor L/\gamma \rfloor$. If $\bar{N} < N_L$ there is nothing to prove since the bounds emerge from $\gamma\bar{N} \le L$ and $|\delta_j| < \varepsilon_{\text{esc}}$. So focus on $\bar{N} \ge N_L$. Here $\bar{Z}(s)$ denotes the mean-dynamics path starting from the state at τ_- and let $\bar{P}_n$ be defined analogously. On the fixed horizon $[0, N_L]$ the realized products track their mean-dynamics counterparts to within a factor $1 + O(K\sqrt{\gamma})$, conditional on $\mathcal{F}_{t_-}$. One can

repeat the same type of fixed-horizon tracking argument as in the proof of Lemma 2(iii), starting from the realized state at release, and it is omitted for brevity. Then,

$$G_{\gamma,L} \equiv \gamma\sigma \sum_{m<N_L} \bar{P}_{m+1}^{-1}\eta_{t_-+m}$$

is a conditional Gaussian approximation to M_{N_L}. Its conditional variance is bounded between fixed positive multiples of the gain, uniformly in $L \geq 1$. The lower bound holds because the weights on the first unit of slow time are at least e^{-a}. The upper bound comes from the geometrically decaying weights $\bar{P}_{m+1}^{-1}$. The normal density bound leads to

$$\Pr\big(|\delta_0 + M_{N_L}| \leq 2r_0\sqrt{\gamma} \mid \mathcal{F}_{t_-}\big) \leq Cr_0 + o(1).$$

For every $\eta > 0$, choose r_0 small and then L large. The claimed sign and scaling event then occurs with probability at least $1 - \eta - o(1)$. Thus, making it unlikely that $|\delta_0 + M_{N_L}|$ remains near zero. As before, Condition P and Doob's inequality leads to

$$\Pr\left(\sup_{N_L \leq n \leq \bar{N}} |M_n - M_{N_L}| > r_0\sqrt{\gamma}\right) \leq C_B r_0^{-2} e^{-2a_\varepsilon L}.$$

Choose r_0 small and then L large to make the probability of this event small. Then, with probability approaching 1,

$$|S_n - S_{N_L}| = |M_n - M_{N_L}| \leq r_0\sqrt{\gamma} \leq \frac{1}{2}|S_{N_L}|$$

Thus, $1/2|S_{N_L}| \leq |S_n| \leq 3/2|S_{N_L}|$. So then $|S_{N_L}| = |\delta_0 + M_{N_L}|$ fluctuates at order $\sqrt{\gamma}$, and so any subsequent discounted noise is small relative to $|S_{N_L}|$.

It remains to sum the path of δ_j. For $n_0 \leq j < \bar{N}$, the S_n bound in (11) implies[2]

$$|\delta_j| = P_j|S_j| = \frac{P_j|S_j|}{P_{\bar{N}-1}|S_{\bar{N}-1}|}|\delta_{\bar{N}-1}| \leq C_S|\delta_{\bar{N}-1}|\frac{P_j}{P_{\bar{N}-1}}.$$

By construction, $|\delta_{\bar{N}-1}| < \varepsilon_{\text{esc}}$. Applying Condition P to $[j, \bar{N}-1]$ implies that this bound becomes

$$|\delta_j| \leq C_S\varepsilon_{\text{esc}}e^B e^{-a_\varepsilon\gamma(\bar{N}-1-j)+o(1)}. \tag{14}$$

Evaluating geometric sums of $\delta_j|$ and δ_j^2 with these bounds proves (12). Three events were not considered along L. Coefficients deviate from the mean-dynamics path more than $K\sqrt{\gamma}$, $|\delta_0 + M_{N_L}| < 2r_0\sqrt{\gamma}$, and $|M_n - M_{N_L}| > r_0\sqrt{\gamma}$. The probabilities of these can be made arbitrarily small by first taking K large, then r_0 small, then L large, and finally γ small. This proves Claim 1.

[2]The initial block $j < n_0$ satisfy bounds $\gamma\sum_{j<N_L}|\delta_j| \leq L\varepsilon_{\text{esc}}$ and $\gamma\sum_{j<N_L}\delta_j^2 \leq L\varepsilon_{\text{esc}}^2$, which are absoreved into $C_b\varepsilon_{\text{esc}}$ and $C_b\varepsilon_{\text{esc}}^2$.

Claim 2: the benchmark's memory of the transition is discounted away at the rate of gain, and the exit path supplies too little variation to refit a trend, and any remaining learning noise vanishes along with the gain. There are finite constants C_{inh} and C_B^{ℓ}, uniform over the compact class of initiating states, such that on the stopped interval

$$\gamma \sum_{j<\bar{N}} \ell^{+}_{t_-+j} \leq C_{\text{inh}} + C_B^{\ell} \varepsilon_{\text{esc}} + o_p(1). \tag{15}$$

The first term is the finite amount of diagnostic evidence following from Phase 2. The second term is the additional evidence generated by the repulsion away from α_M.

In establishing the claim, it is convenient to rewrite the benchmark updating equations as

$$R_{c,t+1} = (1-\gamma)R_{c,t} + \gamma z_{c,t} z'_{c,t},$$
$$s_{c,t+1} = (1-\gamma)s_{c,t} + \gamma z_{c,t} x_t,$$

using the change of variable $s_{c,t} \equiv R_{c,t}\theta_{c,t}$ and denoting $s^M \equiv R^M(\alpha_M, 0)'$. Regularity assumptions make the actual coefficients recoverable as $\theta_{c,t} = R_{c,t}^{-1} s_{c,t}$ (a mapping that is Lipschitz on the projected set). Then we can define the deviations vector $Y_t \equiv \left(R_{c,t} - R^M, s_{c,t} - s^M, \hat{\sigma}^2_{t-1} - \sigma^2, \hat{\sigma}\right.$

At $(\delta, Y) = (0,0)$, which occurs with the system pinned at α_M, the PLM and benchmark predictive densities coincide pointwise. The LLR bound (1) implies

$$\ell^{+}_{t_-+j} \leq L_j\{|\delta_j| + \|Y_j\|\} \qquad \mathbb{E}L_j^2 \leq C.$$

So for the claim to hold requires that

$$\gamma \sum_{j<\bar{N}} \ell^{+}_{t_-+j} \leq \gamma \sum_{j<\bar{N}} L_j|\delta_j| + \gamma \sum_{j<\bar{N}} L_j\|Y_j\|$$

The term $\gamma\sum_{j<\bar{N}} L_j|\delta_j| = \gamma\sum_{j<n_0} L_j|\delta_j| + \gamma\sum_{n_0\leq j<\bar{N}} L_j|\delta_j|$, where $|\delta_j| < \varepsilon_{\text{esc}}$ and with fixed slow-time L, makes the first component order $O_p(\varepsilon_{\text{esc}})$. Then Claim 1 establishes (14) for $j \geq n_0$, so the first term is

$$\gamma \sum_{j<\bar{N}} L_j|\delta_j| \leq C_S \varepsilon_{\text{esc}} e^B \left[\gamma \sum_{j<\bar{N}} e^{-a_\varepsilon \gamma(\bar{N}-1-j)} L_j\right] + o_p(1) \equiv C_S \varepsilon_{\text{esc}} e^B W_{\bar{N}-1} + o_p(1)$$

where $W_{n+1} = \rho_\gamma W_n + \gamma L_{n+1}$, $\rho_\gamma = e^{-a_\varepsilon\gamma} \approx 1 - a_\varepsilon\gamma + O(\gamma^2)$, so W_n is a stable EWA. Since the derivatives of the LLR are (at most) quadratic in (x_j, x_{j-1}), and $x_j = T_{V_j}(c_j) + \sigma\eta_j$, we can choose $L_j \leq C_0\left[1 + \eta_j^2 + \eta_{j-1}^2\right] \equiv \bar{L}_j$. Let $q_\gamma = \lceil \frac{1}{\gamma} \rceil$, which is 1-unit of slow-time that can be divided into blocks $I_r = \{rq_\gamma, \ldots, (r+1)q_\gamma - 1\}$. Then $B_r = \gamma\sum_{j\in I_r} \bar{L}_j$ satisfies $\mathbb{E}B_r = \gamma\sum_{j\in I_r} \mathbb{E}\bar{L}_j \leq C$ and $\text{Var}(B_r) \leq C\gamma^2 q_\gamma \leq C\gamma$. Now pick $\bar{B} = 6C_0$, define $B_L = \bar{B} + d$, and pick $B_L - \mathbb{E}B_r \geq d$. Then $\Pr\left(\mathrm{B_r} > \mathrm{B_L}\right) \leq \frac{\text{Var}(\mathrm{B_r})}{\mathrm{d}^2}$. With 1 block taking $1/\gamma$ periods there are $K_\gamma = O(\log(1/\gamma))$ blocks. Then we can compute a union bound

$$\Pr\left(\sup_{0\leq \mathrm{r} < \mathrm{K}_\gamma} \mathrm{B_r} > \mathrm{B_L}\right) = \Pr\left(\bigcup_{\mathrm{r}=0}^{\mathrm{K}_\gamma - 1} \{\mathrm{B_r} > \mathrm{B_L}\}\right) \leq \frac{\mathrm{C}\gamma}{\mathrm{d}^2}\mathrm{K}_\gamma = \mathrm{O}(\gamma \log(1/\gamma))$$

and it follows that $\sup_{0\le r<K_\gamma} B_r = O_p(1)$. We've seen that $W_n \le \gamma \sum_{j<n} \rho_\gamma^{n-j} \bar{L}_j \le C_{a_\varepsilon} \sup_{r\le K_\gamma} B_r$, and so $\sup_{n<N_T} W_n = O_p(1)$. Noting that $\bar{N} \le N_T$, establishes the bound on the first term $\gamma \sum_{j<\bar{N}} L_j|\delta_j| = O_p(\varepsilon_{\text{esc}} e^B)$.

The rewritten benchmark recursions imply a variation of constants with $C < \infty$ and $\omega > 0$ such that

$$Y_j = \Psi_j Y_0 + \gamma \sum_{i<j} \Psi_{j-1-i} B_i \delta_i + Y_j^{\text{noise}} + Y_j^{\text{rem}}, \qquad \|\Psi_j\| \le C e^{-\omega j \gamma}. \tag{16}$$

The first term integrates as

$$\gamma \sum_{j<\bar{N}} L_j \|\Psi_j Y_0\| \le C_\Psi \|Y_0\| \gamma \sum_{j<\bar{N}} L_j e^{-\omega\gamma j} \le C_\Psi \bar{Y}_- \left\{ \frac{C_L}{\omega} + o_p(1) \right\} \equiv C_{\text{inh}} + o_p(1),$$

where

$$\bar{Y}_- \equiv \sup_{z\in\mathcal{Z}} \|Y(z)\| < \infty, \qquad C_{\text{inh}} \equiv \frac{C_\Psi C_L}{\omega} \bar{Y}_-.$$

The second inequality follows because $\bar{N} \le N_T$, the random Lipschitz coefficient is bounded as above, and

$$\gamma \sum_{j\ge 0} e^{-\omega\gamma j} = \frac{\gamma}{1 - e^{-\omega\gamma}} = \frac{1}{\omega} + O(\gamma),$$

while the corresponding centered weighted sum is $o_p(1)$. The constant C_{inh} is fixed before L is chosen. This ordering is possible because the Lipschitz bound holds uniformly on the (projected) state space. So, C_{inh} collects the contraction rate ω, the L_j bound C_L, and the projection bound $\bar{Y}_-$. The second term integrates like

$$\begin{aligned}
\gamma \sum_{j=1}^{\bar{N}-1} L_j \left\| \gamma \sum_{i=0}^{j-1} \Psi_{j-1-i} B_i \delta_i \right\| &\le \gamma^2 \sum_{j=1}^{\bar{N}-1} L_j \sum_{i=0}^{j-1} \|\Psi_{j-1-i}\| \, \|B_i\| \, |\delta_i| \\
&\le C\gamma^2 \sum_{j=1}^{\bar{N}-1} L_j \sum_{i=0}^{j-1} e^{-\omega\gamma(j-1-i)} \|B_i\| \, |\delta_i| \\
&= C\gamma \sum_{i=0}^{\bar{N}-2} \|B_i\| \, |\delta_i| \left[\gamma \sum_{j=i+1}^{\bar{N}-1} L_j e^{-\omega\gamma(j-1-i)} \right] \equiv C\gamma \sum_{i<\bar{N}-1} \|B_i\| \|\delta_i\| \tilde{W}_i
\end{aligned}$$

Then using Claim 1 $|\delta_i| \le C\varepsilon_{\text{esc}} e^B w_i, w_i = e^{-a_\varepsilon \gamma(\bar{N}-1-i)}$, the last term is bounded by $C\varepsilon_{\text{esc}} e^B \gamma \sum_i w_i \|B_i\| \tilde{W}_i$. The same blocking argument used for W_n establishes that $\sup_i \tilde{W}_i = O_p(1)$, the coefficients B_i have uniformly bounded second omments, and since $\gamma \sum_i w_i = O(1)$, then applying Cauchy-Schwartz shows that the summation is $O_p(1)$, and so this second term is $O_p(\varepsilon_{\text{esc}} e^B)$. Then the third term, utilizing $\mathbb{E}\|Y_j^{\text{noise}}\|^2 \le C\gamma$ and $\mathbb{E}\left(L_j \|Y_j^{\text{noise}}\|\right) \le \left(\mathbb{E} L_j^2\right)^{1/2} \left(\mathbb{E}\|Y_j^{\text{noise}}\|^2\right)^{1/2} \le C\sqrt{\gamma}$ is

$$\mathbb{E}\left[\gamma \sum_{j<N_T} L_j \|Y_j^{\text{noise}}\| \right] \le C\sqrt{\gamma}\gamma N_T = O(\sqrt{\gamma}\log(1/\gamma)) \to 0$$

The last component is the nonlinear remainder. It is quadratic in the local deviations, and so the neighborhood can be chosen so that the terms which include Y_j will only contribute an arbitrarily small multiple of the left-hand side. The remaining δ_j^2 contribution is bounded by Claim 1 as $O_p(\varepsilon_{\text{esc}}^2)$. Thus, we prove Claim 2.

Bounded exit budget. Let $u_t \equiv \ell_t^+$. Iterating on the concern recursion leads to,

$$\gamma \sum_{j<n} \Lambda_{t_-+j-1} = \Lambda_{t_--1} - \Lambda_{t_-+n-1} + \gamma \sum_{j<n} u_{t_-+j}.$$

With $\Lambda \geq 0$ and the projection facility,

$$A_n^+ \leq \frac{3\kappa\overline{\sigma}^2}{k} \left\{ \Lambda_{t_--1} + \gamma \sum_{j<n} u_{t_-+j} \right\}. \tag{17}$$

The definition of τ_- and Assumption 1 imply that $\Lambda_{t_--1} \leq {V^*-\nu+O(\gamma)}/{\kappa\underline{\sigma}^2} \equiv \Lambda_-^{\max}$. Then choose $\eta > 0$ so that

$$\hat{B} = \frac{3\kappa\overline{\sigma}^2}{k}\{\Lambda_-^{\max} + C_{\text{inh}}\} + 3\eta.$$

After fixing $\hat{B}$ and the Claim 1 constants, choose ε_{esc} sufficiently small so that $a_\varepsilon > 0$ and

$$\frac{3\kappa\overline{\sigma}^2}{k} C_{\hat{B}}^{\ell} \varepsilon_{\text{esc}} < \eta.$$

Then let γ be small enough that ${3\kappa\bar{\sigma}^2}/{k}|o_p(1)| < \eta$. Claim 2 implies that

$$A_n^+ \leq \frac{3\kappa\bar{\sigma}^2}{k} \left\{ \Lambda_-^{\max} + C_{\text{inh}} + C_B^{\ell} \varepsilon_{\text{esc}} + o_p(1) \right\}$$

Then we have

$$A_{\bar{N}}^+ \leq \frac{3\kappa\bar{\sigma}^2}{k} \left\{ \Lambda_-^{\max} + C_{\text{inh}} \right\} + 2\eta = \hat{B} - \eta$$

Suppose N_B is the binding stopping time with $B = \hat{B}$. Then $A_{N_B}^+ \geq \hat{B}$ by definition. But this then contradicts $A_{\bar{N}}^+ \leq \hat{B} - \eta$. Thus

$$\Pr(N_B < N_{\text{esc}} \wedge N_T) \longrightarrow 0.$$

Suppose instead, for the high probability event that $N_B > \bar{N}$, that $N_T < N_{\text{esc}}$. Recall that $N_T = \frac{1/2 \log(1/\gamma) + C_T}{a_\varepsilon \gamma} - \theta_\gamma$ for some $\theta_\gamma \in [0, 1)$. Then $a_\varepsilon \gamma N_T = 1/2 \log(1/\gamma) + C_T - a_\varepsilon \gamma \theta_\gamma$. Then using $\sqrt{\gamma} = e^{-1/2 \log(1/\gamma)}$ we arrive at

$$\sqrt{\gamma} e^{a_\varepsilon \gamma N_T} = \exp\left[-\frac{1}{2} \log(1/\gamma) + a_\varepsilon \gamma N_T \right] = \exp[C_T - a_\varepsilon \gamma \theta_\gamma]$$

Now fix C_T so that

$$r_0 e^{C_T - \hat{B}} > 2\varepsilon_{\text{esc}}.$$

then we find the contradiction,

$$|\delta_{N_T}| = P_{N_T}|S_{N_T}| \geq r_0\sqrt{\gamma}\, e^{a_\varepsilon \gamma N_T - \hat{B} - o(1)} = r_0 e^{C_T - \hat{B} - a_\varepsilon \gamma \theta_\gamma - o(1)} > \varepsilon_{\text{esc}},$$

which is inconsistent with $N_T < N_{\text{esc}}$. Therefore

$$\Pr(N_{\text{esc}} < \infty) \longrightarrow 1, \qquad A^+_{N_{\text{esc}}} = O_p(1).$$

The discrete-time budget differs from $A^+_{\text{exit}}(\tau_{\text{esc}} - \tau_-)$ by only $O(\gamma)$ under the slow-time interpolation, so the same conclusion extends to the budget stated in the lemma. Brief recrossings of V^* are not excluded. They can delay the exit, but their cumulative drag is priced by the budget and cannot prevent escape.

Recovering the escape/lingering times. Now we can establish the claim on the cumulative repulsion:

$$\log \frac{\varepsilon_{\text{esc}}}{D_\gamma} - O_p(1) \leq \Phi^*(\tau_{\text{esc}} - \tau_-) \leq \frac{1}{2}\log(1/\gamma) + O_p(1)$$

where $D_\gamma = |\delta_0| \vee \sqrt{\gamma}$ is the effective initial state at τ_-. First the lower bound, which rules out an early, or quick, exit. The stopped time argument showed $\sup_n |M_n| \leq C\sqrt{\gamma}$. We also have for $P_n = \prod_{j<n}(1+\gamma b_j), b_j = \lambda(V_j) - {}^{\delta_j^2}/_k$,

$$\log P_n \leq \gamma \sum_{j<n} \left\{ \lambda(V_j) - \frac{\delta_j^2}{k} \right\} \leq \gamma \sum_{j<n} \lambda(V_j) \equiv \Phi^*(n\gamma) + O(\gamma)$$

So it follows that $|\delta_n| = P_n|\delta_0 + M_n| \leq e^{\Phi^*(n\gamma)} D_\gamma O_p(1)$. At escape, $\varepsilon_{\text{esc}} \leq e^{\Phi^*(\tau_{\text{esc}} - \tau_-)} D_\gamma O_p(1)$, leading to the claimed $\Phi^*(\tau_{\text{esc}} - \tau_-) \geq \log \frac{\varepsilon_{\text{esc}}}{D_\gamma} - O_p(1)$.

Now the upper bound. If escape occurs during the fixed-L interval, the upper bound is immediate. Focus instead on the complementary event. In the period right before escape $|\delta_{N_{\text{esc}}-1}| = P_{N_{\text{esc}}-1}|S_{N_{\text{esc}}-1}| < \varepsilon_{\text{esc}}$, $P_{N_{\text{esc}}-1} < {}^{\varepsilon_{\text{esc}}}/_{r_0\sqrt{\gamma}}, |S_{N_{\text{esc}}-1}| \geq r_0\sqrt{\gamma}$, and also that $1 + \gamma b_{N_{\text{esc}}-1} \leq 1 + \gamma a$ (using $b_j \leq {}^{3V*}/_k \equiv a$) implies $P_{N_{esc}} \leq {}^{\varepsilon_{\text{esc}}}/_{r_0\sqrt{\gamma}}\,(1+\gamma a)$. Then it follows that the cumulative repulsion is bounded by $\log P_{N_{\text{esc}}} \leq {}^1/_2 \log(1/\gamma) + O_p(1)$. Rearrange the definition of $\log P_N$ to get an expression for the linear repulsion Φ^*,

$$\Phi^*(\tau_{\text{esc}} - \tau_-) = \log P_{N_{\text{esc}}} + \frac{\gamma}{k} \sum_{j<N_{\text{esc}}} \delta_j^2 + o_p(1)$$

So we arrive at the bound,

$$\Phi^*(\tau_{\text{esc}} - \tau_-) \leq \frac{1}{2}\log(1/\gamma) + O_p(1)$$

The last step is to convert the bounds on the repulsion rate into bounds on lingering time $\tau_{\text{esc}} - \tau_-$. Recall the identity $\Phi^*(\tau_{\text{esc}} - \tau_-) = a\,(\tau_{\text{esc}} - \tau_-) - A^+_{\text{exit}}\,(\tau_{\text{esc}} - \tau_-)$. The lower bound with $A^+_{\text{exit}} \geq 0$ implies that $a(\tau_{\text{esc}} - \tau_-) \geq \log {}^{\varepsilon_{\text{esc}}}/_{D_\gamma} - O_p(1)$, and so

$$\tau_{\text{esc}} - \tau_- \geq \frac{k}{3V^*} \left\{ \log \frac{\varepsilon_{\text{esc}}}{D_\gamma} - O_p(1) \right\}$$

And for the upper-bound

$$a\left(\tau_{\text{esc}}-\tau_{-}\right)=\Phi^{*}\left(\tau_{\text{esc}}-\tau_{-}\right)+A_{\text{exit}}^{+}\left(\tau_{\text{esc}}-\tau_{-}\right)\leq\frac{1}{2}\log(1/\gamma)+O_{p}(1)$$

which proves the lemma.

□

Proof of Corollary 3. The new ingredient here is to derive a lower bound on the displacement scale at the time of release from the pitchfork neighborhood. Everything else is covered in Theorem 2. The sharper lower bound emerges when $e^{-A^{-}}$ is greater than the $\sqrt{\gamma}$ noise floor.

From the exact factorization (7), the initial state at pitchfork release is $\left(\prod_{t\in[\tau_0,\tau_-}g_t\right)\delta(\tau_0)$. On $[t_0,t_+)$, $V_t>V^*+\nu,\lambda(V_t)=-3/k(V_t-V^*)\equiv-a_t$, and $g_t=1-\gamma\left(a_t+\delta_t^2/k\right)$. While on $[t_+,t_-)$, $|\lambda(V_t)|\leq\lambda^*$ and $g_t\geq1-\gamma\left(\lambda^*+\delta_t^2/k\right)$. Thus, $g_t\geq1-\gamma h_t, h_t\equiv a_t\mathbf{1}_{\{t<t_+\}}+\lambda^a st\mathbf{1}_{\{t_+\leq t<t_-\}}+\delta_t^2/k$. For small γ,

$$\prod_t g_t\geq-\gamma\sum_{t_0\leq t<t_+}a_t-(\tau_--\tau_+)\lambda^*-\frac{\gamma}{k}\sum_{t_0\leq t<t_-}\delta_t^2-O(\gamma)$$

The first term is $\gamma\sum_{t\in[t_0,t_+)}a_t=A^-(\gamma)+O(\gamma)$. The other terms are bounded independently of γ. Thus, $\prod_t g_t\geq c_1e^{-A^-(\gamma)}, c_1=e^{-C_0}>0$. With the separated entry requiring $|\delta_{t_0}|\geq d_{\text{sep}}$, the initial mean-dynamics state after the pitchfork release is at least $c_1d_{\text{sep}}e^{-A^-(\gamma)}$, while the accumulated noise at τ_- is $O_p(\sqrt{\gamma})$ as in parts (i)–(iii) of Lemma 3.

Now we work off the two branches of the initial displacement D_γ. Consider first the budget-limited case $e^{-A^-(\gamma)}\geq K\sqrt{\gamma}$ with K large. The noise cannot cancel this initial state. Since the accumulated noise, normalized by $\sqrt{\gamma}$, is $O_p(1)$, for every $\epsilon>0$ there is K such that, with probability at least $1-\epsilon$,

$$|c(\tau_-)-\alpha_M|\geq\tfrac{1}{2}c_1d_{\text{sep}}e^{-A^-(\gamma)}.$$

Hence $D_\gamma=\Theta_p(e^{-A^-(\gamma)})$ and $\log(\varepsilon_{\text{esc}}/D_\gamma)=A^-(\gamma)+O_p(1)$. The post-release exit-drag budget and cubic occupation are both $O_p(1)$ by Lemma 3(iv), so the actual P_n differs from the largest possible linear counterpart $3V^*(\tau-\tau_-)/k$ by only $O_p(1)$. The exit driven by the initial state will occur when

$$\frac{3V^*}{k}(\tau_{\text{esc}}-\tau_-)=\log\frac{\varepsilon_{\text{esc}}}{D_\gamma}+O_p(1)=A^-(\gamma)+O_p(1).$$

This leads to the reported

$$\tau_{\text{esc}}-\tau_-=\frac{k}{3V^*}A^-(\gamma)+O_p(1)$$

on this first branch. Since $e^{-A^-(\gamma)}\geq K\sqrt{\gamma}$ implies $A^-(\gamma)\leq\frac{1}{2}\log(1/\gamma)-\log K$, this equals $(k/3V^*)\min\{A^-(\gamma),\frac{1}{2}\log(1/\gamma)\}+O_p(1)$.

On the other noise-floor branch $e^{-A^-(\gamma)}<K\sqrt{\gamma}$, the release scale D_γ is at least $\sqrt{\gamma}$ and the result is covered by Corollary 2. Adding that $\tau_--\tau_0$ is bounded, concludes that

$$\tau_{\text{linger}}=\frac{k}{3V^*}\min\Big\{A^-(\gamma),\tfrac{1}{2}\log(1/\gamma)\Big\}+O_p(1).$$

□

4 Collapse-channel reachability

This section gives the full proof of Proposition 3. The paper's appendix records its four steps. The proof covers the speed-ρ family, $\rho > 0$, with critical value $\kappa_C^\nu(\rho) \equiv (V^* + \nu)/(\underline{\sigma}^2 \underline{\ell}_\rho)$. Proposition 3 is the case $\rho = 1$, where $\underline{\ell}_1$ equals the bound $\underline{\ell}$ of Theorem 1.

Proof of Proposition 3. The proof proposes a candidate path for c (the reverse-drift path) that runs from near an outer well into the set of initiating states $\mathcal{I}_C^\nu$, and shows that the shocks needed to implement that path have finite cost. Fix a speed $\rho > 0$ satisfying the analogue for H_ρ of Phase 2's residual-window condition, and choose the endpoint truncation $\varepsilon > 0$ so that the terminal ε-ball lies inside the $\epsilon_0/2$-neighborhood of α_M from Definition 4(ii), denoted B_M below. Let $b_0(c) \equiv G(c) - c$ and let ψ_ρ solve

$$\dot{\psi}_\rho = -\rho b_0(\psi_\rho)$$

between the truncated outer and middle neighborhoods. Since the path is bounded away from the fixed-point endpoints, its duration $T_\rho < \infty$.

For the full three-phase episode, the system starts at the outer quiescent state, transitions into the initiating set, lingers at the sticky threshold, and exits. Since the proposed transition path is truncated at both endpoints, with the terminal endpoint in the initiating set, the first step is to construct a path that connects the quiescent state to the reverse-drift candidate transition path. So the path under consideration here prepends a smooth connector of fixed duration, running from z_+ to the truncated (by ε) outer endpoint. The connector can be chosen inside the quiescent neighborhood, so the residual-window lower bound on the transition segment still holds. The sequence of mean shocks implementing the connector is bounded by projection and compactness and therefore has finite action I_{conn}. We reset the clock at the start of the reverse-drift segment. The augmented vector of coefficients initializes at the connector's terminal position.

The main text shows that the mean observation needed to implement the candidate c-path is

$$\xi_\rho \equiv \psi_\rho + \dot{\psi}_\rho = (1+\rho)\psi_\rho - \rho G(\psi_\rho) = H_\rho(\psi_\rho).$$

The associated law of motion is

$$X_\rho(\tau) = \xi_\rho(\tau) + \sigma\widetilde{\eta}, \qquad \widetilde{\eta} \sim N(0,1).$$

Call $\bar{\zeta}_\rho$ the mean-dynamics path for the augmented coefficients that come from $X_\rho(\tau)$, with initial condition at the connector's terminal state. Then there is the implied mean-path $\bar{V}_\rho = \kappa \bar{\Lambda}_\rho \bar{\widehat{\sigma}}_\rho^2$.

Construct the sequence of mean shocks that leads to the c-path as

$$q_\rho(\tau) \equiv \frac{\xi_\rho(\tau) - T_{\bar{V}_\rho(\tau)}(\psi_\rho(\tau))}{\sigma} = -\frac{(1+\rho)b_0(\psi_\rho(\tau)) + \frac{1}{2}G''(\psi_\rho(\tau))\bar{V}_\rho(\tau)}{\sigma}.$$

Then,

$$T_{\bar{V}_\rho}(\psi_\rho) - \psi_\rho + \sigma q_\rho = -\rho b_0(\psi_\rho) = \dot{\psi}_\rho.$$

This construction shows that $(\psi_\rho, \bar{\zeta}_\rho)$ is a feasible path.

The candidate path is bounded away from the fixed-point endpoints, $\bar{V}_\rho$ is bounded by projection, G'', b_0 are bounded on the compact interval, so $\sup_{\tau<T_\rho} |q_\rho(\tau)| < \infty$. Then we have,

$$I_C^{\text{cand}} = I_{\text{conn}} + \frac{1}{2}\int_0^{T_\rho} q_\rho(\tau)^2\, d\tau < \infty.$$

To complete the assessment of the candidate path it remains to verify that it terminates in $\mathcal{I}_C^\nu$. Since G' is quadratic, as shown earlier H_ρ has no more than three monotonicity intervals. Together with the assumed residual-window condition, the Phase 2 detectability result applies and gives

$$\ell_\rho^*(\psi_\rho) \equiv \int_0^{T_\rho} e^{-(T_\rho - s)} \bar{u}_\rho(s; \psi_\rho)\, ds \geq \underline{\ell}_\rho > 0.$$

For vigilance, note that $\bar{\Lambda}_\rho(T_\rho) = e^{-T_\rho}\bar{\Lambda}_\rho(0) + \int_0^{T_\rho} e^{-(T_\rho - s)}\bar{u}_\rho(s;\psi_\rho)\, ds$ with $\bar{\Lambda}_\rho(0) \geq 0$, so $\bar{\Lambda}_\rho(T_\rho) \geq \ell_\rho^*(\psi_\rho) \geq \underline{\ell}_\rho$. Then with the variance floor and $\kappa > \kappa_C^\nu(\rho)$, $\bar{V}_\rho(T_\rho) \geq \kappa\underline{\sigma}^2\underline{\ell}_\rho > V^* + \nu$. So, $\psi_\rho(T_\rho)$ is inside B_M and $\bar{V}_\rho(T_\rho) > V^* + \nu$, so the terminal state of the candidate path is in the interior of $\mathcal{I}_C^\nu$. Further, by continuity of the mean-dynamics in the initial state, there is a small closed ball $K_C \Subset \mathcal{I}_C^\nu$ around this terminal state such that every state in K_C satisfies Definition 4. From every state in K_C the mean belief path enters B_M within a common bounded slow time, and along the way the mean Jensen intensity stays above $V^* + \nu$ by a common positive clearance. This is the uniformity that Theorem 2 requires of its initiating compact. If the separated-entry refinement of Corollary 3 is desired, the endpoint truncation is chosen so that every point of K_C also satisfies $|c - \alpha_M| \geq d_{\text{sep}}$.

Now compute the action cost. Note that the finite-cost conclusion does not rest on Assumption 3, which just provides an action-cost interpretation. Let q^{cand} be the connector and reverse-drift mean shocks concatenated over their finite total horizon of $N_\gamma = \lfloor (T_{\text{conn}} + T_\rho)/\gamma \rfloor$ periods, write $q_j \equiv q^{\text{cand}}(j\gamma)$, and let Q_γ be the distribution, for fixed initial z, that makes $\widetilde{\eta}_j = \eta_j - q_j \sim N(0,1)$ and let P_γ denote the original distribution. Then the relative entropy is $D(Q_\gamma \| P_\gamma) \equiv E_{Q_\gamma} \log \frac{dQ_\gamma}{dP_\gamma} = \frac{1}{2}\sum_j q_j^2 = \gamma^{-1}\{I_C^{\text{cand}} + o(1)\}$, which includes the $o(1)$ Riemann error. Define a fixed tube around the candidate path as $A_\gamma = \left\{\sup_{j \leq N_\gamma} \|Z_j - \bar{Z}(j\gamma)\| < r\right\}$, where $\bar{Z}$ is the concatenated candidate mean path and the radius r is no larger than the radius of K_C. The truncated skeleton keeps a positive distance from the projection boundary, so the projection does not bind on a small enough tube. Repeating the Phase 2 tracking argument then shows that $Q_\gamma(A_\gamma) \to 1$.

Now using the change-of-measure and a direct likelihood-ratio argument leads to the lower bound on P_γ stated in the proposition. It is useful to expand

$$\log \frac{dQ_\gamma}{dP_\gamma} = \sum_{j=0}^{N_\gamma - 1} q_j \eta_j - \frac{1}{2}\sum_{j=0}^{N_\gamma - 1} q_j^2$$

Recall that $\widetilde{\eta}_j \equiv \eta_j - q_j$ is standard normal under Q_γ, so substituting $\eta_j = q_j + \widetilde{\eta}_j$ gives

$$\log \frac{dQ_\gamma}{dP_\gamma} = \frac{1}{2}\sum_{j=0}^{N_\gamma - 1} q_j^2 + \sum_{j=0}^{N_\gamma - 1} q_j \widetilde{\eta}_j.$$

The first term is $\gamma^{-1}\{I_C^{\text{cand}} + o(1)\}$ as already stated. The second term has mean zero under Q_γ and standard deviation $(\sum_j q_j^2)^{1/2} = O(1/\sqrt{\gamma})$. Now define the event, for any $\delta > 0$, $B_{\gamma,\delta} = \{|\gamma \sum_j q_j \widetilde{\eta}_j| < \delta\}$. Chebyshev's inequality gives $Q_\gamma(B_{\gamma,\delta}) \to 1$. On $B_{\gamma,\delta}$, putting together the bounds, $\log dP_\gamma/dQ_\gamma \geq -(I_C^{\text{cand}} + o(1) + \delta)/\gamma$. The change-of-measure is

$$P_\gamma(A_\gamma) = E_{Q_\gamma}\left[\frac{dP_\gamma}{dQ_\gamma}\mathbf{1}_{A_\gamma}\right] \geq \exp\left\{-\frac{(I_C^{\text{cand}} + \delta + o(1))}{\gamma}\right\} \times Q_\gamma(A_\gamma \cap B_{\gamma,\delta})$$

With $Q_\gamma(A_\gamma), Q_\gamma(B_{\gamma,\delta}) \to 1$, it follows that $\liminf_{\gamma\to 0} \gamma \log P_\gamma(A_\gamma) \geq -I_C^{\text{cand}} - \delta$, and since $\delta > 0$ is arbitrary, $\liminf_{\gamma\to 0} \gamma \log P_\gamma(A_\gamma) \geq -I_C^{\text{cand}}$. On A_γ the terminal state Z_{N_γ} lies in K_C. Let $\mathcal{C}_\gamma$ be the event that, initializing in K_C, the collapse-and-linger episode runs and beliefs exit the ε_{esc}-neighborhood. Lemma 3 and Theorem 2 establish that $\inf_{z'\in K_C} \Pr_{z'}(\mathcal{C}_\gamma) = 1 - o(1)$. With $\mathcal{E}_\gamma$ denoting the full three-phase event running to exit, combining with the Markov property at the deterministic time N_γ,

$$P_\gamma(\mathcal{E}_\gamma) \geq \mathbb{E}_{P_\gamma}\left[\mathbf{1}_{A_\gamma} \Pr_{Z_{N_\gamma}}(\mathcal{C}_\gamma)\right] \geq P_\gamma(A_\gamma) \inf_{z'\in K_C} \Pr_{z'}(\mathcal{C}_\gamma)$$

Then since $\gamma \log \inf_{z'\in K_C} \Pr_{z'}(\mathcal{C}_\gamma) \to 0$, the stated bound follows as $\liminf_{\gamma\to 0} \gamma \log P_\gamma(\mathcal{E}_\gamma) \geq -I_C^{\text{cand}}$. Finally, this bound extends uniformly over a small compact neighborhood K_o of z_+ by fixing the same sequence of candidate mean shocks for every starting state. The tracking bound propagates a fixed initial displacement with a constant that depends only on the horizon, so choosing the diameter of K_o small enough keeps the tube event likely, $\inf_{z\in K_o} Q_\gamma(A_\gamma) \to 1$. Then the bound is uniform, $\liminf_{\gamma\to 0} \gamma \log \inf_{z\in K_o} \Pr_z(\mathcal{E}_\gamma) \geq -I_C^{\text{cand}}$.

In standard learning for this environment (i.e. $V \equiv 0$) the speed-ρ reverse-drift candidate path produces an action proportional to $(1+\rho)^2/\rho$, minimized at $\rho = 1$. This is consistent with the least-cost paths in Williams (2026, Proposition 3). The present proposition only needed to specify one feasible speed and does not make any least-cost claims for the full augmented system. □

5 How does vigilance impact escapes?

While the paper emphasizes the sticky thresholds, it is reasonable to conjecture that vigilance accelerates all escapes from an outer well. The intuition comes from truncating concern to the accumulated positive part of the LLR, which is first order in the benchmark's first-order autocorrelation advantage even though the (untruncated) mean LLR is second order. Concern updating only occurs on the favorable half of the LLR fluctuations. A small persistent perturbation of the innovations raises vigilance and the flattened landscape lowers the exit barrier, either to the threshold or the other well. Turning this into a theorem requires asymptotics for the full augmented system and detectability estimates both of which lie outside the paper's fixed-V large-deviation approach. We, therefore, leave to future work calculating the relative likelihood of the collapse and basin-switching channels.

6 Further details on the EHR illustration

The growth-cycle application in Section 5.2 is EHR-inspired rather than an exact replica of Evans et al. (1998). The model was translated into the form analyzed here by first taking EHR's Euler and technology loci using their reported parameterization and investment-cost schedule. Over the interval bounded by their numerical low- and high-growth paths, the calibration here produces balanced-growth rates

$$1.00260, \qquad 1.02585, \qquad 1.04872,$$

with a slope of 3.589 at the E-unstable middle path. The exact EHR map is locally convex at both outer paths and doesn't feature the inward, sign-changing curvature used in the sticky-threshold mechanism. Instead, EHR disciplines the locations of the balanced-growth paths and the slope at the middle path, but not the global curvature of the temporary-equilibrium equation.

Specifically, the normalized temporary-equilibrium map is

$$G_N(g) = g - \frac{(g+1)(g-.5)(g-2)}{k_N}, \qquad k_N = \frac{2.25}{3.589-1} = .8687. \tag{18}$$

The affine transformation

$$g^{\text{gross}} = 1.0256588 + .01537253(g - .5)$$

maps the two outer roots to the low- and high-growth paths above and places the normalized middle root within two basis points of the middle EHR path. Thus, we reproduce the economically meaningful equilibrium values and slope at the middle point while also imposing the cubic curvature that makes the vigilance mechanism transparent.

We assume that the shock follows a stationary AR(1)

$$\varepsilon_t = .6\varepsilon_{t-1} + .16\eta_t.$$

With a persistent state, the PLM is expanded to an AR(1) $g_t = c + b\,(g_{t-1} - c) + e_t$. For numerical implementation, we project g_t on

$$z^P_{t-1} = (1, \widetilde{g}_{t-1})', \qquad \widetilde{g}_{t-1} = (g_{t-1} - .5)/1.5,$$

with coefficients $\theta = (\theta_0, \theta_1)'$. The paper's Table 1 reports the SCEE coordinates $b = \theta_1/1.5$ and $c = (\theta_0 - .5\,b)/(1-b)$. Fitted values, residual variances, and the LLR are the same in either parameterization. Previously the diagnostic benchmark leveraged its ability to detect serial correlation and now we let the benchmark detect non-linearity by adding the second- and third-order Legendre terms $P_2(\widetilde{g}_{t-1})$ and $P_3(\widetilde{g}_{t-1})$. The benchmark nests the PLM and can detect the nonlinear variation omitted by the restricted forecasting class.

Computing the persistent RPE proceeds as follows. For a candidate pair (θ, V), hold the PLM coefficients and Jensen intensity fixed and simulate the stationary distribution induced by the ALM

$$g_t = G_N(m_t) + \frac{1}{2}G''_N(m_t)V + \varepsilon_t, \qquad m_t = \theta_0 + \theta_1\,\frac{\theta' z^P_{t-1} - .5}{1.5},$$

in which m_t is the paper's "t-1"-timed forecast, $m_t = c + b^2(g_{t-1} - c)$. Under the resulting distribution, let $\theta^P(\theta, V)$ and $\sigma_P^2(\theta, V)$ denote the population one-step projection of g_t on z_{t-1}^P and its residual variance, and let $\Lambda(\theta, V) = \mathbb{E}[(\ell_t^*)^+]$ denote the associated population vigilance. A population vigilance RPE solves

$$\theta^P(\theta, V) = \theta, \qquad \kappa\Lambda(\theta, V)\sigma_P^2(\theta, V) = V. \tag{19}$$

Local stability is assessed from the Jacobian of the full mean recursion for the PLM, benchmark, moment matrices, variance estimates, and vigilance, rather than from the scalar slope of G_N.

Table 1 reports on the computed equilibrium coefficient values. Denote the residual variances for the PLM and benchmark, respectively, as $\sigma^{2*} = \sigma_P^2(\theta^*, V^*)$ and σ_c^{2*}. Stochastic simulations indicate local stability.

References

Evans, George W. and Seppo Honkapohja, *Learning and Expectations in Macroeconomics*, Princeton, New Jersey: Princeton University Press, 2001.

_ , _ , and Paul Romer, "Growth Cycles," *American Economic Review*, 1998, *88*, 495–515.

Williams, Noah, "Equilibria, Folds, and Escapes," 2026. Working paper, Miami Herbert Business School.